

Fernando Agustin Teodoro

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SUMMARY

Melbourne-based software developer and tinkerer who likes understanding each piece of the puzzle and how it affects the whole. I build full-stack apps end to end, from data models and APIs to mobile-first UI, tests and cloud deploys. I use AI-assisted development with a review-first workflow, and some of my projects were built entirely by hand.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, C#, Dart, SQL

Front end & mobile: React, React Native (Expo), Flutter, SwiftUI/WidgetKit, Tailwind, Unity

Back end & data: Node.js (Meteor), Django, Flask, REST APIs, PostgreSQL/PostGIS, MongoDB, SQLite

Real-time: WebSockets (Meteor pub/sub, Supabase Realtime, a hand-written relay), GTFS Realtime

Delivery & systems: Git, GitHub Actions, Docker, Google Cloud (Run, Build, Scheduler), YAML, Linux

Testing: Mocha/Chai, node:test, unittest, Django tests; unit, integration and regression

AI-assisted development: Claude Code, Claude Design, Copilot; I own the plans and review every change

Ways of working: Agile/Scrum, pair programming, technical docs, root-cause analysis

SOFTWARE PROJECTS

Realtime Queue Management System

Jan 2026 – Present

React · TypeScript · Node.js (Meteor) · MongoDB · Mocha/Chai · GitHub Actions · Docker · Cloud Run

- Co-designed the multi-tenant, five-role access model with a teammate, then built its follow-on work
- Built the wait-time engine as a per-provider simulation, then merged three drifting copies of the calculation into one, fixing wrong estimates and closing a cross-facility data leak
- Traced a slow, staggered list load to an N+1 pattern (about three subscriptions per row); replaced it with one batched subscription plus indexes where loads had been full collection scans
- Made actions feel instant on weak mobile networks with optimistic UI that rolls back on failure
- Grew the suite to 900+ unit and integration tests, run in GitHub Actions on pull requests into the release branch; each release auto-deploys to Cloud Run
- Built the foundations by hand from Jan; since Aug I use Claude Code and review every change

Pa-Ride (Carpool Marketplace)

Aug 2026 – Present

React Native (Expo) · TypeScript · Supabase (PostgreSQL, PostGIS, Realtime) · Claude Code

- Researched how Filipinos already carpool in Facebook groups before designing the product and schema
- Built PostGIS corridor matching so riders find drivers already passing near their route
- Made seat booking row-locked Postgres functions behind row-level security, so trips can't be overbooked
- Built shareable trip cards and public trip pages so drivers' Facebook posts pull new users into the app

Sim Racing Telemetry Dashboard (Gran Turismo 7)

Sep 2026

Python · JavaScript · UDP · WebSockets · Canvas/SVG · node:test / unittest · Claude Code

- Built a GT7 dashboard: a Python bridge relays the PS4's UDP telemetry to the browser over WebSockets
- Traced lap times drifting from the game's own to frames timestamped on arrival over jittery Wi-Fi; switching to the game's clock cut lap-time error from 384 ms to 9 ms
- Measured each network hop to find the console's Wi-Fi stalling about 100 ms twice a second, and added a tunable jitter buffer that removed the visible stutter
- Wrote 166 automated tests, including replays of real recorded sessions and a regression test per bug fix

Melbourne Public Transport Widget

Mar – Dec 2025

Flutter · SwiftUI/WidgetKit · Python (Flask) · MongoDB · GTFS Realtime · Google Cloud

- With a partner, built a Flutter app and iOS home-screen widget with live departures for saved routes
- Implemented PTV's HMAC-SHA1 API signing and merged it with GTFS Realtime feeds into one data model
- Built a nightly Cloud Run scraper that rebuilds the ~130 MB timetable only when PTV's data changes

Portfolio Site

Aug 2026 – Present

Python · HTML/CSS/JavaScript · Markdown/YAML · Pages CMS · GitHub Actions · Claude Code

- Built the site on a small Python static site generator, auto-deployed to GitHub Pages on each change
- Set up a Git-based CMS so content and resumes can be edited in a browser, live in about a minute
- Built the Lab as a home for live demos, with each tool kept in its own repo and pulled in at build time

C-LARA Image Annotation Tool (University Capstone)

Aug – Nov 2023

Python · Django REST Framework · React · Agile/Scrum

- As Product Owner in a 5-person Scrum team, turned the client's evolving spec into sprint requirements
- Built the Django REST API (11 endpoints) and its 37-test suite; the project was graded H1

Space Golf (Procedural 3D Golf Game)

Sep – Oct 2022

C# · Unity · 4-person university team · built without AI

- Built procedural planet generation from Perlin noise, so no planet needed hand-modelled terrain
- Wrote a custom Unity Editor inspector so the team could tweak terrain without re-running the game

COVID-19 Air Quality Analysis

2021

Python · Pandas · NumPy · scikit-learn · built without AI

- Led a 3-person team; requested EPA Victoria's data directly and joined it with three other datasets
- Chose Mutual Information over linear methods; found no significant link between mobility and air quality

WORK EXPERIENCE

Freelance QA Tester | UK ETA Visa Services (iOS), via Upwork

May 2026

- Tested the full visa flow end to end instead of the brief's 10 to 15 minute pass, tracing how inputs affected later screens; earned a 5-star review

Ad-hoc Programmer | NutTAG Australia

Dec 2023 – Jan 2024

- Got up to speed on a product NutTAG had just inherited, helping the team understand it
- Redesigned the support chatbot's flow and knowledge base to target complaints in NutTAG's reviews

TINKERING

- Bridged ISP routers to my own mesh and gaming routers at home, tuning DNS, Wi-Fi channels, QoS and port forwarding for steadier connections
- Revived a failing laptop with Pop!_OS Linux and fixed its drivers
- Set up smart homes (Google Home/Nest, Alexa) on their own network; built and tuned my own sim rig

COMMUNITY

Australian Computer Society | member and event volunteer, 30+ CPD hours logged in 2026 2025 – Present

Google Developer Group Melbourne | attendee

2024 – Present

EDUCATION

BSc Computing & Software Systems | University of Melbourne

2023

- Coursework: Foundations of Algorithms (C), Computer Systems (C, socket programming), Object-Oriented Software Development (Java), Software Modelling and Design (Java)

International Baccalaureate Diploma | German-European School Manila

2018

ADDITIONAL

Languages: English (native), Filipino (conversational), German and Italian (basic)

Other: Typing 130 wpm at 97% accuracy · Victorian RSA · Front of house at Royal Stacks (2022)